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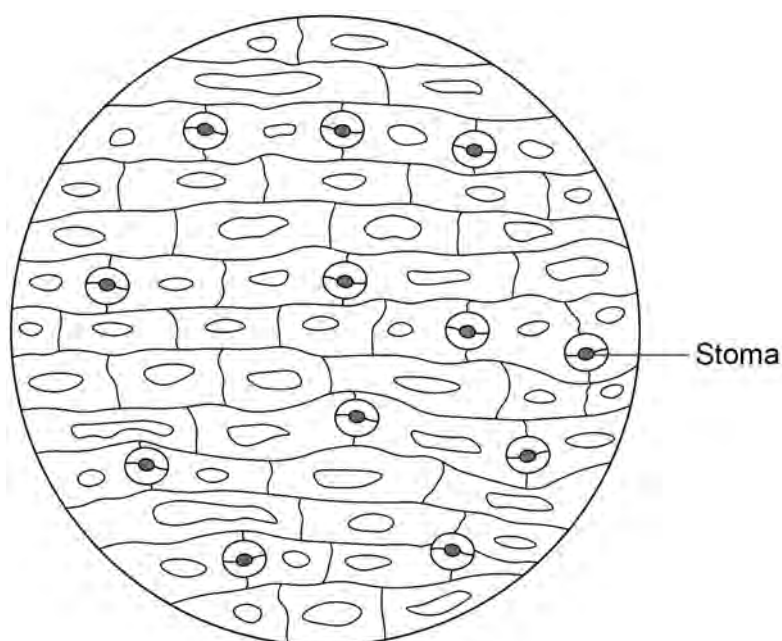
A student investigated the distribution of stomata on a leaf.

The student:

- observed one area of the lower surface of a leaf using an optical microscope
- counted all the stomata.

Figure 7 shows the area of the leaf the student observed.

Figure 7



The diameter of the field of view was 0.36 mm

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Use **Figure 7** and the information provided to calculate the stomatal density of the lower leaf surface.

The area of a circle can be calculated using the formula πr^2

For π use 3.14

Give your answer to 2 significant figures.

[3 marks]

Answer _____ stomata mm⁻²



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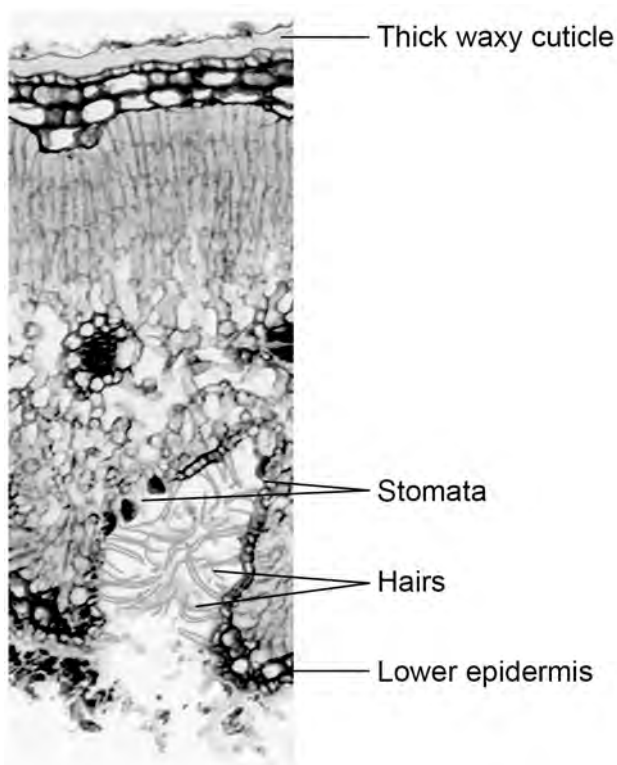
Suggest **one** improvement the student could make to obtain a more reliable value for stomatal density.

[1 mark]

0 4 . 3

Figure 8 shows a transverse section of a leaf from a xerophytic plant.

Figure 8



Explain how **one** feature shown in **Figure 8** reduces water loss from the leaf.

[1 mark]

Feature _____

How it reduces water loss _____

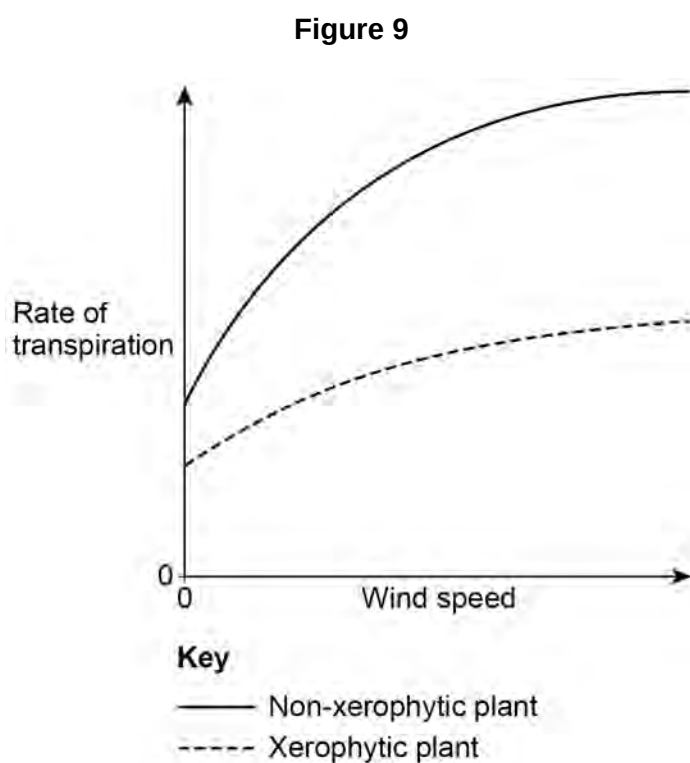
Question 4 continues on the next page

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Figure 9 shows the effect of wind speed on the rate of transpiration of a xerophytic plant and a non-xerophytic plant.

Do not write
outside the
box



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Give **two** conclusions that can be made from **Figure 9**.

[2 marks]

1 _____

2 _____

